Chapter 5: The Fundamental Unit of Life

5.1 What are Living Organisms Made Up Of? In this chapter, we shall discuss the basic structural unit of an organ, which is the cell. Cells were first discovered by Robert Hooke in 1665 while examining a thin slice of cork through a self-designed microscope. He observed small box-like structures and called them cells.

The cell is the fundamental structural and functional unit of all living organisms. All living organisms are made up of cells. Some organisms are made up of a single cell (unicellular), while others are made up of many cells (multicellular).

5.2 What is a Cell Made Up Of? The cell has three main parts: the cell membrane, the nucleus, and the cytoplasm.

The cell membrane or plasma membrane is the outermost covering of the cell. It separates the contents of the cell from its external environment. The cell membrane is selectively permeable, meaning it allows some substances to pass through while blocking others.

The process of movement of substances from a region of higher concentration to a region of lower concentration is called diffusion. Osmosis is the movement of water molecules through a selectively permeable membrane from a region of higher water concentration to a region of lower water concentration.

5.3 Cell Organelles Every cell has a membrane around it to keep its own contents separate from the external environment. Large and important parts of the cell are called organelles.

The Nucleus: The nucleus is the most important organelle of the cell. It contains the genetic material (DNA) and controls all cellular activities. The nuclear membrane has pores that allow the transfer of material between the nucleus and cytoplasm.

Mitochondria: Mitochondria are known as the powerhouses of the cell. They are the sites of cellular respiration, where food (glucose) is broken down to release energy in the form of ATP. Mitochondria have their own DNA and ribosomes, which suggests they were once independent organisms.

Endoplasmic Reticulum (ER): The ER is a large network of membrane-bound tubes and sheets. There are two types: - Rough ER: has ribosomes on its surface and helps in protein synthesis - Smooth ER: helps in the manufacture of fat molecules and detoxification

Golgi Apparatus: The Golgi apparatus consists of a system of membrane-bound vesicles arranged approximately parallel to each other. Its functions include storage, modification, and packaging of cell products.

Lysosomes: Lysosomes are membrane-bound sacs filled with digestive enzymes. They break down foreign materials and worn-out cell organelles. When the cell gets damaged, lysosomes may burst and the enzymes released digest their own cell. Therefore, lysosomes are also known as the suicide bags of the cell.